

ATC Testing

Advanced Trigonometry Calculator Testing

This document explains the current test infrastructure for Advanced Trigonometry Calculator (ATC).

Regression Test Runner

The main automated regression runner is:

```
tests\run-atc-regression.ps1
```

Run it after building ATC:

```
powershell -ExecutionPolicy Bypass -File .\tests\run-atc-regression.ps1 -AtcExe .\x64\Release\atc.exe
```

For x86 builds:

```
powershell -ExecutionPolicy Bypass -File .\tests\run-atc-regression.ps1 -AtcExe .\Release\atc.exe
```

Current validated result for both Release x64 and Release x86:

```
Summary: 359 passed, 0 failed
```

Latest Release x86 and Release x64 builds completed successfully with 0 Warning(s), 0 Error(s), and both executables passed the same regression suite.

Memory Stress Runner

The memory stress runner is:

```
tests\run-atc-memory-stress.ps1
```

Example:

```
powershell -ExecutionPolicy Bypass -File .\tests\run-atc-memory-stress.ps1 -Iterations 10
```

It currently focuses on repeated polynomial simplification, roots-to-polynomial, and textual/coefficient equation solver commands.

Isolated Coverage Helper

The isolated coverage helper is:

```
tests\run-atc-isolated-coverage.ps1
```

Use it when a specific area needs focused validation outside the full regression run.

Current isolated coverage result:

```
Summary: 65 passed, 0 failed
```

Current Automated Coverage

The regression suite currently covers:

- arithmetic basics and precedence;
- constants and precision formatting;
- trigonometry and inverse trigonometry;
- hyperbolic and inverse hyperbolic functions;
- logarithms and custom guide syntax;
- complex-number paths used by documented commands;
- matrix/list functions, matrix helper commands, direct matrix/vector indexing and persisted indexed matrix updates;

- variable names and implicit multiplication;
- sorting and ASCII sorting flows;
- roots-to-polynomial report export;
- polynomial simplification;
- equation solving and systems of equations;
- solver paths including selected trigonometric, hyperbolic, polynomial, and rational-cancellation cases;
- function study examples;
- graph settings, deterministic graph navigation simulation, navigation edge clamping and explicit graph-range navigation;
- date/time commands that can be automated safely;
- TXT detector / command bridge commands, including redirected CMD input, mocked shell/window side effects, and real-file solve txt processing with mocked answer-file opening;
- file/folder command existence checks;
- app environment commands that can run safely under redirected input, including about and auto adjust window;
- variable/result management commands;
- persistent settings and menu retry behavior;
- one safe runtime smoke path for each deep interactive module: unit conversions, microeconomics, physics, statistics, geometry, financial calculations, triangles rectangles solver, and arithmetic matrix solver;
- verbose-resolution behavior;
- interactive prompt behavior, including Tab autocomplete vocabulary, ambiguous-match cycling and Up/Down history navigation;
- double and Boost mp_float precision mode persistence.

The detailed test matrix is maintained in:

`tests\ATC_AUTOMATED_TEST_CASES.md`

Documentation to Test Traceability

Documented behavior should have automated coverage whenever it is deterministic and safe to execute under redirected input.

Use these rules when updating documentation:

- stable command examples should be added to the regression suite where practical;
- interactive modules should have at least a safe smoke test or an explicit manual-validation note;
- commands that open windows, files, browsers or PC-control actions should use mocks, source-level checks or manual validation notes;
- do not update the documented test count unless the suite has actually been executed;
- update `tests\ATC_USER_GUIDE_COVERAGE.md` when a documented area gains or loses coverage.

The Cookbook and Best Practices documents are intended to use already documented commands. If a recipe cannot be tested safely, it should say so through notes or limitations.

Known Manual or Interactive Validation Gaps

The automated suite intentionally avoids or only partially covers:

- shutdown, restart, hibernate, sleep, lock, and similar PC-control commands;
- full-duration clock, stopwatch, timer, and continuously running views;
- commands that open external browsers or editors, except for TXT/file flows

that now have explicit mock coverage;

- full live manual graph rendering and key-buffer behavior;
- remaining branches inside deeply interactive modules such as triangles,

arithmetic matrix solver, conversions, finance, physics, microeconomics, and statistics menus.

Long-running time command branches, short stopwatch marking, zero-duration clock rendering, and timer syntax guards are covered by the isolated helper. Full-duration countdowns, alarms, and continuously running views should still be validated manually or covered later with safe, non-destructive automation.

The direct TXT workflow is covered with temporary fixtures: `predefine txt` writes a real input path, `solve txt` generates the `_answers.txt` response file, invalid lines are kept isolated from later lines, and the answer-file open is recorded through the test mock instead of launching Notepad. The fixture also covers `solve equation` and `simplify polynomial` inside the TXT input, including their export-prompt-safe non-interactive path. Live `auto solve txt` watcher behavior remains an isolated/source-level check until it has a deterministic multi-process fixture.

Settings Modified by Tests

The regression runner may temporarily touch ATC setting files under:

`%USERPROFILE%\Pictures\Advanced Trigonometry Calculator`

The runner backs up and restores the settings it modifies.

Recommended Validation Before Changes

For most changes:

```
powershell -ExecutionPolicy Bypass -File .\tests\run-atc-regression.ps1 -AtcExe .\x64\Release\atc.exe
```

For memory-sensitive work, also run the memory stress runner when practical.